

Formalization of DP with Precedence and Setups

Given a set of jobs $J = \{1, 2, \dots, n\}$, setup costs e_{ij} , and a set of precedence rules P (where $(p \rightarrow j) \in P$ means p must be finished before j can start).

1. State Definitions (X)

We define the state space X_k as the collection of all valid subsets of size k :

- X_k : Subsets of J with cardinality k that satisfy all internal precedence constraints.
- X_0 : The initial state (empty set).

2. Backward Recursion (Step $k = n - 1$ down to 1)

Let $G_o(k, i)$ be the optimal cost to complete the remaining jobs starting from the state i in level k .

For each subset i in level k , and each subset j in level $k + 1$:

- Let $\text{job_next} = X_{k+1}(j) \setminus X_k(i)$.
- **Transition Rule:** The transition $i \rightarrow j$ is valid if and only if:
 1. $X_k(i) \subset X_{k+1}(j)$
 2. $\text{Predecessors}(\text{job_next}) \subseteq X_k(i)$

The cost transition is defined as:

$$G(k, i, j) = \begin{cases} e_{\text{last}(i), \text{job_next}} + G_o(k + 1, j) & \text{if valid} \\ \infty & \text{otherwise} \end{cases} \quad (1)$$

The optimal value for the state is:

$$G_o(k, i) = \min_j G(k, i, j) \quad (2)$$

3. Initial Step (Passo 0)

The final optimization over the first job chosen:

$$G_0(j) = e_{\text{start}, X_1(j)} + G_o(1, j) \quad \forall j \text{ s.t. } \text{Pred}(X_1(j)) = \emptyset \quad (3)$$